Problem Statement

Modern System-on-Chip (SoC) designs consume a significant amount of leakage power even when functional blocks remain idle. Conventional power-gating techniques typically rely on static ON/OFF policies, which either keep a block fully powered or completely shut it down. Although effective in reducing leakage, these approaches often introduce unnecessary wake-up overhead and fail to efficiently handle varying workload patterns. Therefore, there is a need for a lightweight adaptive power-management mechanism capable of reducing leakage while maintaining acceptable performance and hardware overhead.

Motivation

As technology scales into deep-submicron nodes, leakage power becomes an increasingly dominant component of total chip power. Many embedded processors and digital accelerators spend a considerable amount of time in idle or partially active states, making static power management inefficient. A lightweight adaptive controller that dynamically selects different sleep modes can improve the trade-off between leakage power and wake-up latency without requiring complex prediction algorithms. This motivates the development of lightweight hardware-based adaptive techniques that can intelligently manage power states with minimal implementation overhead.

Related Work Snapshot

Prior adaptive power-management work falls into three categories relevant to this project: (1) ML/clustering-based multi-core workload prediction (Karthikeyan et al. — K-means + LSTM combined with DVFS and core-level gating), which achieves strong savings but carries prediction-hardware overhead unsuitable for a single small block; (2) fixed-threshold multi-level MTCMOS gating (Agarwal et al. — three sleep depths selected by static idle-cycle counts), which is lightweight but prone to thrashing under bursty workloads because it ignores recent activity density; and (3) wake-up current-shaping techniques (Chang et al. — two-phase footer activation), which address the transition between states but provide no decision logic for when to transition. No prior work in this space combines a dual-condition (idle-count and activity-rate) decision gate with adaptive hysteresis-band widening, while remaining cheap enough (shift-register + comparators, zero multipliers) for a single ALU-class block.

Research Gap

Fixed-threshold adaptive power-gating controllers decide sleep entry from idle-cycle count alone. This is demonstrably insufficient under bursty traffic: a workload with short, frequent idle gaps will cross a fixed idle threshold repeatedly, causing the controller to enter and exit sleep on every gap (thrashing), which costs more in wake-up energy and latency than it saves in leakage. No published lightweight (non-ML) controller combines an activity-rate signal with idle-count to suppress this failure mode.

Research Objective

The objective of this work is to design and evaluate a lightweight adaptive multi-level power-gating controller for a small digital block. The controller monitors hardware activity and dynamically transitions between ON, LIGHT SLEEP, and DEEP SLEEP states using a lightweight activity-history-based hysteresis decision mechanism. The proposed design is evaluated through RTL implementation, functional verification, synthesis, place-and-route, and power analysis using the Synopsys design flow.

Scope

This work focuses on RTL-level design and evaluation of the power-management controller. The functional block selected for evaluation is a 32-bit ALU. Software-based power management, operating-system scheduling, and dynamic voltage/frequency scaling (DVFS) are outside the scope of this project.

Baseline Definition

To support a fair quantitative comparison, a static ON/OFF power-gating controller will be implemented as the baseline: it gates the same alu_32bit block using a single fixed idle-cycle threshold (no activity-rate term, no hysteresis band, no adaptive widening), modeled after the fixed-threshold approach in Agarwal et al.'s multi-level MTCMOS gating. The baseline and the adaptive design will share the same ALU RTL, the same isolation/retention strategy, the same UPF power domains, and will be evaluated under identical synthesis constraints and identical synthetic workloads, varying only the decision logic under comparison.

Fair-Comparison Protocol

1. Same technology library (generic `typical.db` or whichever PDK is finalized) for both designs, all corners.
2. Same clock constraint set per design point — both designs synthesized at both the low-power (15 ns) and high-speed (8 ns) corners; never compare a result from one design point against the other's different corner.
3. Same operating voltage / supply assumptions in UPF for both designs.
4. Same `alu_32bit.v`, same isolation strategy (ISO_AND, clamp-to-0), same retention assumption for any state-retentive sleep mode.
5. Same testbench workloads (burst, sparse, mixed, sustained-idle) applied to both designs, byte-for-byte identical stimulus.
6. Metrics reported per design, per corner: leakage power, dynamic power, area, Fmax/critical path, wake-up latency, and number of mode transitions (thrashing indicator).

Expected Contributions

1. A lightweight adaptive multi-level power-gating controller for small RTL-based digital systems.
2. A lightweight activity-monitoring mechanism for estimating block utilization.
3. A hysteresis-based decision policy that reduces unnecessary sleep-state oscillations.
4. A complete RTL implementation validated using the Synopsys RTL-to-GDSII design flow.
5. A quantitative comparison against a conventional ON/OFF power-gating controller using identical implementation constraints.

Assumptions

See Fair-Comparison Protocol above for the full list of controlled variables.